Stocker — an AI Investment Team on n8n

Final project · **Track 1 — Financial Automation: a Multi-Agent n8n System**

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Repository: github.com/gilbarel1/algotrade-project · August 2026

▶ Watch the 5-minute demo

6	4	3	148	$0.02–0.07
n8n workflows 159 nodes	specialist agents 3 run in parallel	Risk Manager passes draft → critique → final	offline tests green in CI	per run logged per agent

1 · The financial problem

Forming a view on a stock means synthesizing three unlike sources: price action, news flow, and exchange disclosures. Doing that by hand across a watchlist is slow and inconsistent. Doing it with a single LLM prompt is fast but untrustworthy — one-shot pipelines invent financial figures, hide their uncertainty, and overcommit on thin evidence.

Tel-Aviv names add specific friction. Coverage splits across Hebrew and English, and TASE disclosures live on a bot-protected JavaScript site whose actual figures sit in PDF attachments two layers below the page a naive scraper would read.

What this project solves. Given a watchlist that may mix TA-35 and S&P 500 names, produce a justified **long / short / hold / avoid** recommendation per ticker — with a conviction level, a complete reasoning trace, and citations back to every source. Two guarantees hold throughout: **no financial figure is ever fabricated**, and **no uncertainty is ever hidden**. A failed data source lowers conviction; it never becomes a signal.

2 · What was built

A virtual investment team of four AI agents orchestrated by **n8n**, backed by a local **FastAPI** service that owns all real computation. Three specialists gather independent evidence per ticker; a Risk Manager weighs it through a three-pass self-critique and issues the call. Every run produces a **PDF report** with an executive summary, per-ticker pages, and the full reasoning trace.

Agent	Role	Key mechanism
Sentiment claude-haiku-4.5	Reads recent news from NewsAPI and RSS, English and Hebrew	**Dual scoring.** An LLM and a fine-tuned transformer (FinBERT for English, DictaBERT for Hebrew) score every article independently. Their disagreement is itself a reported signal.
Earnings grok-4.3	Reads exchange disclosures — Maya for TASE, SEC EDGAR for US, routed by market	**Self-consistency.** Every figure is sampled three times at temperature 0.3 and committed only on a majority vote of verbatim matches. Anything else is marked `ambiguous`.
Technical gemini-2.5-flash-lite	Narrates RSI, MACD, Bollinger and ATR computed server-side	The model narrates pre-computed JSON only. No number in the report originates inside a language model.
Risk Manager claude-haiku-4.5, ×3	Synthesizes the three signals into the final call	**Draft → devil's-advocate critique → final**, then a deterministic rubric clamp so a model that breaks the decision rules cannot reach the report.

Three entry points share one pipeline — no duplicated code path:

Entry point	Behaviour
Manual trigger	The whole watchlist on demand, from the n8n editor. Never gated by market hours.
Hourly schedule	

	A per-market gate analyzes only tickers whose market is open: 11:00 Israel time runs the TASE names, 20:00 runs the US ones — each market carrying its own calendar, hours, news feeds, earnings source and currency.
Chat assistant	*"what do you think about Teva?"* runs the real pipeline for `TEVA.TA` in about a minute. Deliberately a **router, not an analyst** — five nodes and a single tool, so it can only invoke the pipeline and relay what comes back.

3 · Architecture

Two layers joined by one HTTP boundary. n8n holds the orchestration and every LLM call, and contains no machine-learning code. The FastAPI service owns everything needing a real library — indicators (`pandas-ta`), transformer inference, PDF rendering (WeasyPrint over Jinja2), and scraping (Playwright for Maya's protected SPA, plain HTTP for EDGAR) — persisting to **DuckDB**: prices, news, earnings, runs, recommendations, costs.

Each run opens with a run id and the watchlist, fans the three analysis agents out per ticker at concurrency 3, runs the Risk Manager's three passes per ticker, harvests real token usage from n8n's execution API, renders the PDF, and closes the run.

- **Heavy data never crosses the LLM boundary.** Price arrays, article bodies and rendered HTML stay server-side; models see short text and scores.
- **Every LLM response is schema-validated.** A malformed reply earns one stricter retry, then an explicit degraded result — never a silent fallback.
- **Failures degrade, they never crash.** Yahoo, NewsAPI, RSS, Maya, EDGAR and the n8n API each fail into a labelled state that the Risk Manager turns into lower conviction.
- **Costs are observable.** Every call is logged with tokens, dollars and latency; a five-ticker run costs roughly **$0.04–0.08**.
- Times are stored in UTC and rendered in Asia/Jerusalem. Secrets live only in a gitignored `.env`.

4 · The AI, beyond a baseline pipeline

These are not five separate tricks but one principle at three scales: **an LLM cannot be trusted to report its own uncertainty, so uncertainty is derived from disagreement between independent attempts** — never from stated confidence, which HeBERT put at 0.998 while being wrong (§5.1). Samples disagreeing ⇒ a figure becomes `ambiguous`; two models disagreeing ⇒ conviction is capped; two passes disagreeing ⇒ the call is downgraded. The system never asks "how sure are you?": confidence is earned by surviving a challenge, and §5.2 measures whether the challenges have teeth.

Technique	What it adds
Dual sentiment LLM alongside FinBERT / DictaBERT	An independent, non-LLM opinion on every article. Disagreement above a threshold caps conviction at medium and is printed in the report — model conflict becomes information rather than something averaged away.
Self-consistency extraction n = 3, temperature 0.3, majority vote	"Never invent numbers" enforced by construction rather than by instruction. A figure is committed only when repeated samples agree on text present verbatim in the source; everything else prints as `ambiguous`, visually distinct in the PDF. Sampling makes it stochastic, so it sharply reduces invented figures and makes the residue *measurable* — see §5.1.
Three-pass critique loop with a deterministic rubric clamp	The recommendation visibly stress-tests itself, and all three passes are printed. A mechanical clamp then enforces the decision rubric — and when it overrides the model, the report says so rather than quietly rewriting the model's words.
Versioned few-shot prompts with an evaluation harness	Prompts are files, not workflow internals: version-controlled, reviewable, and scored against hand-labelled fixtures by a single command.
Degraded-agent epistemics	A failed agent is defined as *no information*. The prompts state that an empty result caused by a system failure says nothing about the company, and the rubric counts it for neither side while still lowering conviction.

5 · Results

5.1 · Evaluation harness

Scored against hand-labelled fixtures: 30 news items (20 English, 10 Hebrew), 10 disclosures, and 7 chat-router cases.

Agent	Result
Sentiment — LLM	accuracy **0.90** (27 / 30) · MAE 0.12
Sentiment — transformers, overall	accuracy **0.77** (23 / 30) · MAE 0.39
└ English — FinBERT	accuracy **0.80** (16 / 20) · MAE 0.43
└ Hebrew — DictaBERT	accuracy **0.70** (7 / 10) · MAE 0.32
Sentiment — agreement	Pearson r **0.82** (n = 30)
Earnings — classifier	macro-F1 **1.00** · materiality accuracy 0.90 (n = 10)
Earnings — extractor	precision **1.00** · recall **1.00** · **ambiguous-when-absent 22 / 22**
Chat router	refusal **4 / 4** · routing **3 / 3** · no fabrication

The transformer arm is reported per language, because it is two models and the mean of the two describes neither. Reaching 0.70 on Hebrew meant replacing the model — and the investigation is recorded because two of its steps failed:

Step	Result
HeBERT — the original choice	**0.30.** Not weakly discriminating but *not discriminating at all*: **neutral for 10 of 10** Hebrew items at 0.833–0.998 confidence, including "record quarterly orders, guidance raised" at 0.998.
Two rescue attempts	Re-normalising polarity over the polar classes made it **worse — 0.53** (the residual mass is noise, not signal); a threshold sweep ±0.1 → ±0.6 moved nothing beyond a single item.
DictaBERT — adopted	**0.70.** Every negative and every neutral read correctly; it misses only positives.

Agreement between the arms rose **0.70 → 0.82** as a direct consequence: a model outputting ~0 for everything cannot correlate with anything, so the earlier figure was largely measuring noise.

The extractor row is the "never invent numbers" guarantee, measured — including its limit. Because self-consistency *samples* (three times at temperature 0.3), it is not deterministic: four measured runs scored `ambiguous`-when-absent at 22/22, 21/22, 22/22 and 22/22 — the single miss being a figure committed that its source does not state. The honest claim is that the mechanism sharply reduces invented figures and makes the residue measurable — not that it eliminates them. A single-sample extractor offers neither guarantee nor a way to count its own failures.

These figures **reproduce**: re-run eleven days later, every metric returned identical — only the cost moved, by 676 tokens. That is what pinned fixtures and `temperature = 0` outside the sampler are for; the one number that *should* wander is the extractor's, and it is reported rather than smoothed.

5.2 · Ablations — do the techniques earn their cost?

Every technique above costs real tokens, and "we used self-consistency" is a claim about effort, not effect. A second harness (`npm run ablations`) switches each one off and re-scores.

Technique removed	Measured consequence
Self-consistency n = 3 vote → n = 1 sample	Invented figures **0 → 2**; invented rate **0% → 3%**; precision **1.00 → 0.92**. Per draw: 5%, 0%, 5% — two of three samples each committed a figure absent from the source, and **the vote caught both because the samples disagreed with one another**. Disagreement between draws *is* the detector; one sample has no access to it.
Few-shot examples 9 examples → none	Accuracy **0.90 → 0.77**, MAE **0.12 → 0.20**. The 0.77 is the point: it is exactly what the *free* local transformer scores, so without the examples the paid model stops being worth paying for.
The critique loop final → draft	Over 26 recommendations the call changed **8× (31%)**, *every one* `long → hold`; conviction **9× (35%)**, **9 downgrades, 0 upgrades**. It has **never once** made the system

more bullish or confident — nor is it a rubber stamp: in 19 of 26 cases it argued for a different call.

Both self-consistency arms score the *same* three draws, so the comparison is paired and costs no more than one extraction run; the critique arm needs no LLM calls, the `draft` being already the counterfactual. **What this does not show:** samples are small (10 disclosures, 30 items, 26 recommendations); the critique arm is observational, its draft written by a prompt that *knows* a critique is coming; and "changed the call" is not "improved the call" — nothing here measures whether the `hold` was right, which is what the missing backtest (§7) would test. Method and full caveats: `docs/ablations.md`.

5.3 · A silent data-integrity bug, and what finding it changed

The worst defect here raised no error and failed no test. It surfaced by reading `/ohlc` output: **every Sunday bar was an exact copy of the previous Thursday's**. TASE trades Sunday–Thursday, so cleaning reindexed onto that week — a rule taken from configuration. But **Yahoo returns `.TA` bars Monday–Friday with no Sunday sessions at all** (verified on three TASE names). Reindexing therefore **discarded every real Friday** and **forward-filled a synthetic Sunday**: on `TEVA.TA`/180 days, **35 fabricated bars ≈ 19% of the series** and **31 real sessions dropped**. Each phantom has a zero return and near-zero true range, **deflating ATR** and **flattening RSI**. After the fix: zero fabricated bars, every Friday kept, `AAPL` byte-identical.

The fix removes the assumption rather than correcting it: the grid is derived from the weekdays the feed actually delivers, so no bar can be invented and none discarded, for any market or future source — and a mismatch is reported per symbol rather than absorbed. Two lessons outrank the fix. First, it contradicted the system's own central rule — a pipeline built to refuse inventing an earnings figure was quietly inventing a fifth of its price history, because **guardrails hold where they are enforced, not where they are stated**. Second, **the bug outlived its own repair**: insert-or-replace can correct or add a row but never *retract* one, so stored phantoms kept feeding the endpoints until a re-ingest was made authoritative for its window. Both are now pinned by tests. What those Friday bars *are* remains **unknown** (§6, §7) — rather than guess, the code makes the narrower claim that holds either way.

5.4 · Live behaviour, including under failure

A single-ticker chat run takes about a minute for two to four cents. The reference run behind the committed sample, `r_2026-08-16T12:53`, covered both markets in one pass via the chat assistant — **245 s**, **$0.0686**, all three specialists `ok` — verified against the persisted rows rather than the chat reply, and rendering one report with per-market currencies and a market-grouped summary. On the US side the extractor committed **$109.4 billion** and **$2.02** from Apple's 8-K with all three samples agreeing, and refused `guidance` as `ambiguous` because the exhibit does not state it: the never-invent rule doing both of its jobs on live data rather than a fixture.

> **Case study — what happens when a data source dies**
>
> With a rejected news API key, sentiment degrades to zero articles. In an early run the critique pass argued that the *absence* of news was "itself a material anomaly… both scenarios bearish" — inferring a market signal from an infrastructure failure. Root cause: the agent's failure status was not machine-readable to the prompts. After the fix the final call states that *"a degraded agent is neutral by definition… its absence is not a bearish or bullish signal"*, while still lowering conviction. Failure, diagnosis and before/after evidence are in the history.

5.5 · The reasoning trace, as printed in the report

Excerpts from three passes for `NFLX`, after the fix above; each is printed in full on the ticker's page.

1 · DRAFT — hold · low

"Agreement count is 1 of 3… Per rule 2, one or fewer agents agreeing mandates a 'hold' call regardless of conviction cap."

2 · DEVIL'S-ADVOCATE CRITIQUE — counter: hold

"MACD below signal line — a subtle bullish inflection the draft dismisses as neutral… the denominator includes a degraded agent that contributed zero information. **The system correctly avoids inventing sentiment signal**, but the true agreement ratio is understated."

3 · FINAL DECISION — hold · low

"The critique's observations do not override the mechanical agreement rule… **per the rubric, a degraded agent is neutral by definition and does not constitute evidence of direction — its absence is not a bearish or bullish signal**."

The critique attacks the draft on its own evidence; the final answers each objection rather than restating the draft. Full reports: `docs/samples/`.

6 · Risks and caveats

- **The critique loop reduces overconfidence; it does not guarantee correctness.** It is a structured reasoning aid, not a financial-validity guarantee — and the report's footer says so.
- **No live execution and no return measurement.** The system argues positions; it never checks whether they would have made money.
- **Maya scraping is best-effort.** A protected JavaScript site rendered headless, with figures two layers down in a PDF attachment. A scanned file or dead link yields `ambiguous` figures — never guessed ones.
- **EDGAR extraction is a bounded excerpt, not a structured parse.** A 10-Q's tables flatten badly into text, so the selection logic explicitly prefers press-release exhibits.
- **News coverage is uneven.** Measured live: `TEVA.TA` returns 0 items in the default 3-day window against 47–49 for large US names. Auto-derived search terms also fail where a registered name differs from the press name ("Alphabet" vs "Google", hand-curated).
- **A model swap is not a no-op.** Models clear schema validation at different rates and the retry path is a different branch — swapping the earnings model surfaced a latent fan-in bug the previous one had masked.
- **One open item.** The agreement rule counts a degraded agent in its three-agent denominator, so "1 of 3" may really be "1 of 2 measurable" — raised by the system's own critique pass during a live run.
- **Yahoo's Tel Aviv calendar does not match the real trading week, and the reason is unknown** (§5.3). Ingestion follows the feed's own calendar and reports the mismatch, so it invents no bar and discards none — but whether those Friday bars are real, mislabelled, or an artefact is unresolved. TA-35 reports generated before the fix understate their indicators; the committed sample was regenerated after it.
- **A guardrail is only as good as its enforcement point.** Never-invent was mechanised for earnings figures and merely assumed for price data — exactly where fabrication appeared. That generalises to anything here not pinned by a test.
- **Thin TA-35 sentiment is genuine, not broken.** In the reference run `TEVA.TA` returned zero articles, both scores 0. The count is printed and nothing padded — but a dual-sentiment split cannot inform a call with no news to score.

7 · Future development

- **Backtest the recommendation stream** against realized returns — turning rationale quality into measured alpha, net of costs.
- **Delivery channels.** E-mail and Telegram are each a single additional n8n node; the report step is already structured for them.
- **Scale the watchlist** to the full TA-35 and S&P 500. The per-market architecture supports it today; a paid news tier is the binding constraint.
- **Rubric refinements.** Resolve the degraded-denominator question, and consider a stronger Risk Manager model — a one-field change, with the cost table as arbiter.
- **OCR fallback** for scanned disclosure PDFs, currently an honest `ambiguous`.
- **Settle the TASE calendar question (§5.3)** against the exchange's own session data or a second vendor. Current behaviour is safe, but "we don't know what those Friday bars are" is a gap in provenance, not a design choice.

Running it. `npm run setup`, keys into `.env`, `npm run dev` (quant service + n8n + chat UI); the six workflows are imported once in the n8n editor. Quick-start and wiring: `README.md`. Design: `docs/design.md`. Run walk-throughs: `docs/results.md`. Ablation method and caveats: `docs/ablations.md`. Demo outline: `docs/demo_script.md`. This PDF is built from its HTML by `npm run summary`, so the two cannot drift.